

Ideas for experimenting with Generative AI: Use cases and things to keep in mind

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The University supports responsible experimentation with Generative AI tools. This page provides an overview of possible use cases, and important things to keep in mind before you begin experimenting. It is intended as general information only and is not a comprehensive list of all the ways in which people across the University are exploring the use of AI. You can find more resources, including available tools, at HUIT's Generative AI page.

Before you experiment with Generative AI tools

- **Review the University's initial guidelines** for use of Generative AI tools, including guidelines around the use of confidential data with publicly-available tools, adhering to your School's student and faculty policies around the use of Generative AI, and your responsibility to review and verify any AI-generated content that you publish or share with others.
- **Explore available tools** through HUIT, including a comparison of the appropriate levels of data that they are approved to handle. The University is working to provide access to more tools, so check back periodically to see if new options are available. Some Schools may also offer additional tools to eligible affiliates.
- **Consider AI's limitations:** A good rule of thumb is to retain a healthy skepticism and understand that these tools can and frequently do get things wrong. AI-generated content can be inaccurate, misleading, or entirely fabricated (sometimes called "hallucinations"). A chatbot can provide a confident-sounding, plausible response to a question that is nonetheless completely false or significantly distorted. Always verify the accuracy of AI-generated output.
- **Be mindful of copyright issues:** AI-generated content can contain copyrighted material and per University guidelines, you are responsible for any AI-generated content that you publish or make available to others. Don't upload content that infringes the intellectual property or other legal rights of any third party, and don't ask LLMs to rewrite or reproduce copyrighted material.

Ideas for experimenting with Generative AI

Creating new content

AI can create stories, poems, music, images, recipes, travel guides, study guides, lesson plans, surveys and much more in seconds from simple, natural language prompts. It can generate new ideas or help you get creatively unstuck. The key to successful experimentation with text-based AI content generation is learning how to use and improve your prompts. You can find some ideas and tips for using prompts here.

Generating or debugging code

Generative AI tools can generate computer code, identify and fix bugs, and enhance code quality and efficiency using natural language prompts, which means that people who don't already know any programming languages can experiment with code. But research has shown that AI-generated computer code can be more likely to contain security vulnerabilities. As with all AI-generated content, review your code before you publish – you are responsible for any content that you publish that includes AI-generated material.

Creating spreadsheet formulas

Learning how to create formulas for Excel and other spreadsheet tools can take time. Generative AI tools can help make this easier for non-experts by enabling you to create formulas using natural language prompts. So even if you don't know the formula you need, or where to find it, you can try asking an AI to, for example, "Write an Excel formula that calculates 10% of the sum of columns B and C," and then copy and paste that formula into your spreadsheet. Again, be sure to double-check for accuracy.

Search

Because of the vastness of data that LLMs have been trained on, Generative AI tools can quickly sift through information from many different sources. Many people use Generative AI as an advanced search tool, and popular search engines such as Google and Bing are increasingly incorporating AI-generated results into their products. However, if the data on which AI has been trained is outdated, the output you receive will be inaccurate. As an example, OpenAI's GPT-3.5 was trained on data up to September 2021, which means answers to general knowledge questions (e.g. "Who is the President of Harvard?") will only reflect its knowledge of information up to that date. Similarly, if an AI model is trained on data that contains biases, unfair stereotypes, etc., these biases can be reflected in its outputs. Learn more about bias in AI and other ethical considerations in this article from Harvard Business Review.

Summarizing documents

Generative AI tools can analyze and generate concise summaries of long documents. This can be used for tasks such as highlighting key insights from research papers, summarizing emails, reports, and meeting notes, or identifying key findings from spreadsheet data. Keep in mind that AI may miss key context or misinterpret findings.

Synthesizing information, including survey responses

Generative AI tools can analyze complex datasets and synthesize the information within. A good example is survey responses: if you have results from a survey that include a lot of open text responses from multiple participants, you can feed those into an AI tool and get an overall summary of what people are saying or pick out the most common keywords and phrases.

Translation

Generative AI tools can quickly translate long passages of text into other languages. As with other uses of AI, exercise caution: it has been reported that while Generative AI tools are good at translating into English, they are not as good at translating English into another language.

Reviewing or enhancing your work

In addition to creating new content, AI can be a helpful partner in editing and enhancing your writing. You can paste in an email or document that you're working on and ask the AI to suggest improvements, look for errors, assess flow and readability, and make suggestions of additional content. However, per University guidelines, don't upload any confidential data into publicly-available Generative AI tools.

Boosting productivity

AI tools can summarize meeting notes, highlight key takeaways, and generate action items. These uses will become more widespread as common productivity tools begin to incorporate AI-powered features directly into their apps.

Improving accessibility

Thoughtful application of AI tools can aid in making content more accessible by identifying problems in your documents, suggesting improvements such as alt-text and audio descriptions for images, or converting speech to text for captioning. AI-powered accessibility checkers can be a valuable time saver, but they can't catch every error. You should still be sure to manually test your content for accessibility. It can be overwhelming to assess all the various tools that are available, but Harvard's Digital Accessibility Services can help you navigate the landscape.

See also: AI